Discrete Mathematics

CSE 4203

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4.3 Primes and Greatest Common Divisors

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Prime Numbers: Definition

Definition

A **prime number** is a positive integer greater than 1 whose only positive divisors are 1 and itself.

Example

Examples of prime numbers: $2, 3, 5, 7, 11, 13, 17, \ldots$

Definition

A **composite number** is a positive integer greater than 1 that is not prime; that is, it has positive divisors other than 1 and itself.

Example

$4, 6, 8, 9, 10, 12, \ldots$ are composite numbers.

Theorem 1 (Fundamental Theorem of Arithmetic)

Every integer greater than 1 can be written uniquely as a prime or as the product of two or more primes, where the prime factors are written in order of nondecreasing size.

Example 2: Prime Factorizations

- $100 = 2 \cdot 2 \cdot 5 \cdot 5 = 2^2 \cdot 5^2$
- $641 = 641$ (prime)
- $999 = 3 \cdot 3 \cdot 3 \cdot 37 = 3^3 \cdot 37$
- $1024 = 2 \cdot 2 \cdot \ldots \cdot 2$ (10 times) $= 2^{10}$

Theorem 2: Trial Division with Example

Theorem 2

If n is a composite integer, then n has a prime divisor less than or equal to $\sqrt{n}$.

Proof:

- If n is composite, it has a factor a such that $1 < a < n$.
- Then $n = a \cdot b$, where $b > 1$ is an integer.
- Suppose $a > \sqrt{n}$ and $b > \sqrt{n}$.
- Then $ab > \sqrt{n} \cdot \sqrt{n} = n$, a contradiction.
- Therefore, $a \leq \sqrt{n}$ or $b \leq \sqrt{n}$.
- Hence, n has a divisor $\leq \sqrt{n}$ (and thus a prime divisor $\leq \sqrt{n}$).

Example: Is 221 prime?

- $\sqrt{221} \approx 14.86 \Rightarrow$ check primes ≤ 13: $2, 3, 5, 7, 11, 13$.
- $221 \div 13 = 17 \Rightarrow 221 = 13 \times 17$, so 221 is composite.

Sieve of Eratosthenes: Finding Prime Numbers

What is it?

The Sieve of Eratosthenes is an ancient algorithm used to find all prime numbers up to a given limit n. It is based on the principle of eliminating the multiples of each prime starting from 2.

Why is it Important?

- Efficient method to generate prime numbers.
- Useful in cryptography, number theory, and algorithms.

Steps of the Algorithm

1. Write all numbers from 2 to n.
2. Start with $p = 2$, the smallest prime.
3. Eliminate all multiples of p greater than p.
4. Move to the next uncrossed number and repeat until $p^2 > n$.
5. Remaining numbers are primes.

The Sieve of Eratosthenes (Animated)

Goal: Find all primes ≤ 50

Steps

1. Start with numbers 2 to 50.
2. Cross out multiples of 2, then 3, then 5, then 7.
3. Remaining numbers are primes.

2	3		5		7				11
	13				17		19		
	23						29		31
					37				41
	43				47				

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~~12~~	13	~~14~~		~~16~~	17	~~18~~	19	~~20~~	
~~22~~	23	~~24~~		~~26~~		~~28~~	29	~~30~~	31
~~32~~		~~34~~		~~36~~	37	~~38~~		~~40~~	41
~~42~~	43	~~44~~		~~46~~	47	~~48~~		~~50~~	

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$\cancel{12}$	13	$\cancel{14}$	$\cancel{15}$	$\cancel{16}$	17	$\cancel{18}$	19	$\cancel{20}$	$\cancel{21}$
$\cancel{22}$	23	$\cancel{24}$	$\cancel{25}$	$\cancel{26}$	$\cancel{27}$	$\cancel{28}$	29	$\cancel{30}$	31
$\cancel{32}$	$\cancel{33}$	$\cancel{34}$	$\cancel{35}$	$\cancel{36}$	37	$\cancel{38}$	$\cancel{39}$	$\cancel{40}$	41
$\cancel{42}$	43	$\cancel{44}$	$\cancel{45}$	$\cancel{46}$	47	$\cancel{48}$	$\cancel{49}$	$\cancel{50}$	

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Primes: 2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47

The Infinitude of Primes

Theorem: There are infinitely many prime numbers.

Proof (by Contradiction):

1. Assume finitely many primes: $p_1, p_2, \ldots, p_n$.
2. Let $Q = p_1 p_2 \cdots p_n + 1$.
3. By Fundamental Theorem of Arithmetic:
    - Q is prime, or
    - Q has a prime factor.
4. None of p_j divides Q ($Q \bmod p_j = 1$).
5. $\Rightarrow$ New prime exists $\rightarrow$ Contradiction.

Conclusion: Infinitely many primes exist.

Intuitive Idea:

- Start with some primes: $2, 3, 5, 7$.
- Multiply: $2 \cdot 3 \cdot 5 \cdot 7 = 210$.
- Add 1: $Q = 210 + 1 = 211$.
- 211 not divisible by $2, 3, 5, 7$.
- So Q is:
    - A new prime, or
    - Has new prime factors.

Idea: We can always create a number requiring a new prime!

Mersenne Primes

Definition: A *Mersenne prime* is a prime of the form:

$$M_p = 2^p - 1, \text{ where } p \text{ is prime.}$$

Why Important?

- Extremely efficient Lucas-Lehmer test for $2^p - 1$.
- Largest known primes are usually Mersenne primes.

Examples:

$$2^2 - 1 = 3, \quad 2^3 - 1 = 7, \quad 2^5 - 1 = 31, \quad 2^7 - 1 = 127$$

But $2^{11} - 1 = 2047 = 23 \times 89$ (not prime).

Progress:

- As of 2018: 50 Mersenne primes known, 19 found since 1990.
- Largest known: $2^{77,232,917} - 1$ (23,249,425 digits), discovered in Dec 2017.

The Prime Number Theorem

Theorem: If $\pi(x)$ is the number of primes $\leq x$, then:

$$\pi(x) \sim \frac{x}{\ln x} \quad \text{as } x \to \infty$$

Meaning:

- $\pi(x)$ approximates to $x/\ln x$ for large x.
- Prime density decreases as x grows.

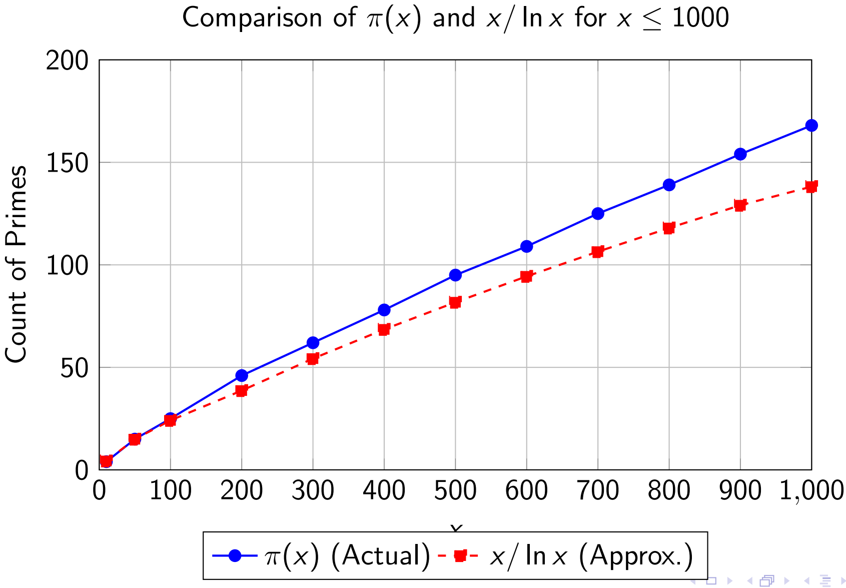
Examples:

$$x = 1000 : \pi(x) = 168, \ \frac{x}{\ln x} \approx 144$$

$$x = 10^6 : \pi(x) \approx 78{,}498, \ \frac{x}{\ln x} \approx 72{,}382$$

Key Idea: There are infinitely many primes, but they get rarer as numbers grow.

Graph: $\pi(x)$ vs $x/\ln x$ (Approximation)



Goldbach's Conjecture

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Related Results:

- Every even integer > 2 is the sum of at most 6 primes (O. Ramaré, 1995).
- Every large integer $=$ prime $+$ (prime or semiprime) (J. R. Chen, 1966).

Twin Prime Conjecture

Definition: Twin primes are pairs of primes that differ by 2:

$$(3, 5),\ (5, 7),\ (11, 13),\ (17, 19),\ (4967, 4969),\ \ldots$$

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Record Twin Primes (as of 2018):

$$2{,}996{,}863{,}034{,}895 \times 2^{1{,}290{,}000} \pm 1$$

(388,342 decimal digits!)

Goal: Find a function $f(n)$ such that $f(n)$ is prime for all positive integers n.

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Example Attempt: Consider the polynomial:

$$f(n) = n^2 - n + 41$$

Observation:

- For $1 \leq n \leq 40$, $f(n)$ is prime.
- Examples: $f(1) = 41,\ f(2) = 43,\ f(3) = 47,\ f(4) = 53.$

Question: Does this hold for all positive integers n?

Observation: The conjecture that $f(n) = n^2 - n + 41$ is prime for all positive integers n is **false**. However, many such simple-sounding problems remain open.

Open Problems in Prime Number Theory

Observation: The conjecture that $f(n) = n^2 - n + 41$ is prime for all positive integers n is **false**. However, many such simple-sounding problems remain open.

Why Important?

- Number theory is full of easy-to-state problems that resist solutions.
- Some conjectures have remained unresolved for centuries.

Even in the 21st century, many fundamental prime-related questions remain unanswered.

Greatest Common Divisor (GCD)

GCD

Definition: Let a and b be integers, not both zero. The greatest common divisor of a and b, denoted by:

$$\gcd(a, b) = \text{largest integer } d \text{ such that } d \mid a \text{ and } d \mid b.$$

Example: Find $\gcd(24, 36)$.

$$\text{Common divisors: } \{1, 2, 3, 4, 6, 12\} \quad \Rightarrow \gcd(24, 36) = 12.$$

Relative Prime

Definition: a and b are *relatively prime* if $\gcd(a, b) = 1$.

Example: $\gcd(17, 22) = 1 \;\Rightarrow$ 17 and 22 are relatively prime.

Pairwise Relatively Prime Integers

Definition

Integers $a_1, a_2, \ldots, a_n$ are *pairwise relatively prime* if:

$$\gcd(a_i, a_j) = 1 \quad \text{whenever } 1 \leq i < j \leq n.$$

Example 1: Check if $10, 17, 21$ are pairwise relatively prime:

$$\gcd(10, 17) = 1, \ \gcd(10, 21) = 1, \ \gcd(17, 21) = 1$$

$\Rightarrow$ Yes, they are pairwise relatively prime.

Example 2: Check if $10, 19, 24$ are pairwise relatively prime:

$$\gcd(10, 19) = 1, \ \gcd(10, 24) = 2 > 1$$

$\Rightarrow$ No, they are not pairwise relatively prime.

GCD Using Prime Factorization

Theorem:

If

$$a = p_1^{a_1} p_2^{a_2} \dots p_n^{a_n}, \qquad b = p_1^{b_1} p_2^{b_2} \dots p_n^{b_n}$$

then:

$$\gcd(a, b) = p_1^{\min(a_1,b_1)} p_2^{\min(a_2,b_2)} \dots p_n^{\min(a_n,b_n)}.$$

Example: Find $\gcd(360, 48)$.

$$360 = 2^3 \cdot 3^2 \cdot 5^1,$$
$$48 = 2^4 \cdot 3^1.$$

Take minimum exponents:

$$\gcd(360, 48) = 2^{\min(3,4)} \cdot 3^{\min(2,1)} \cdot 5^{\min(1,0)} = 2^3 \cdot 3^1 = 24.$$

Least Common Multiple (LCM)

Definition:

The least common multiple (LCM) of positive integers a and b is the smallest positive integer divisible by both a and b. It is denoted by:

$$\operatorname{lcm}(a, b).$$

Formula using Prime Factorization:

If

$$a = p_1^{a_1} p_2^{a_2} \dots p_n^{a_n}, \quad b = p_1^{b_1} p_2^{b_2} \dots p_n^{b_n}$$

then:

$$\operatorname{lcm}(a, b) = p_1^{\max(a_1,b_1)} p_2^{\max(a_2,b_2)} \dots p_n^{\max(a_n,b_n)}.$$

Example: Find $\operatorname{lcm}(360, 48)$.

$$360 = 2^3 \cdot 3^2 \cdot 5^1,$$
$$48 = 2^4 \cdot 3^1.$$

Take maximum exponents:

Relationship Between GCD and LCM

Theorem 5:

For any positive integers a and b:

$$a \cdot b = \gcd(a, b) \cdot \operatorname{lcm}(a, b).$$

Example: Let $a = 24,\ b = 36$.

$$\gcd(24, 36) = 12,\ \operatorname{lcm}(24, 36) = 72.$$

Check:

$$a \cdot b = 24 \cdot 36 = 864, \qquad \gcd(a, b) \cdot \operatorname{lcm}(a, b) = 12 \cdot 72 = 864.$$

Conclusion: The formula holds.

The Euclidean Algorithm

Definition:

The Euclidean Algorithm is an efficient method for finding the greatest common divisor (GCD) of two integers a and b ($a > b > 0$).

Steps:

1. Divide a by b: $a = bq + r$, where $0 \leq r < b$.
2. If $r = 0$, then $\gcd(a, b) = b$.
3. Otherwise, set $a \leftarrow b$, $b \leftarrow r$, and repeat.

Key Property:

$$\gcd(a, b) = \gcd(b, r).$$

Proof of Euclidean Algorithm

Theorem: For integers a, b with $a = bq + r$,

$$\gcd(a, b) = \gcd(b, r).$$

Proof:

- Suppose $d \mid a$ and $d \mid b$. Then $d \mid a - bq = r$, so d divides r.
- Conversely, if $d \mid b$ and $d \mid r$, then $d \mid bq + r = a$.

$$\Rightarrow \text{Common divisors of } (a, b) \text{ and } (b, r) \text{ are identical.}$$

Thus:

$$\gcd(a, b) = \gcd(b, r).$$

Example: Compute $\gcd(252, 105)$

Step 1: Divide 252 by 105:

$$252 = 105 \times 2 + 42.$$

So, $\gcd(252, 105) = \gcd(105, 42)$.

Step 2: Divide 105 by 42:

$$105 = 42 \times 2 + 21.$$

So, $\gcd(105, 42) = \gcd(42, 21)$.

Step 3: Divide 42 by 21:

$$42 = 21 \times 2 + 0.$$

Result: $\gcd(252, 105) = 21$.

Bézout's Theorem and Linear Combination

Theorem 6 (Bézout's Theorem)

If a and b are positive integers, then there exist integers s and t such that

$$\gcd(a, b) = sa + tb.$$

Linear Combination: The greatest common divisor of a and b can be expressed as an integer linear combination of a and b, meaning it can be written in the form

$$sa + tb,$$

where $s, t \in \mathbb{Z}$.

Definition: The integers s and t are called *Bézout coefficients*. The equation is called *Bézout's identity*.

Example:

$$\gcd(6, 14) = 2 = (-2) \times 6 + 1 \times 14.$$

Expressing $\gcd(252, 198)$ as a Linear Combination

Euclidean Algorithm Steps:

$$252 = 198 \times 1 + 54$$

$$198 = 54 \times 3 + 36$$

$$54 = 36 \times 1 + 18$$

$$36 = 18 \times 2 + 0$$

Backward Substitution:

$$18 = 54 - 36 \times 1$$

Substitute $36 = 198 - 54 \times 3$:

$$18 = 54 - (198 - 54 \times 3) = 54 \times 4 - 198$$

Substitute $54 = 252 - 198 \times 1$:

$$18 = (252 - 198) \times 4 - 198 = 252 \times 4 - 198 \times 5$$

Result:

$$18 = 4 \times 252 - 5 \times 198$$

Extended Euclidean Algorithm: Single-Pass Method

Idea: Unlike the two-pass method, the extended Euclidean algorithm computes the Bézout coefficients s and t *during* the Euclidean algorithm by updating sequences s_j and t_j in a single pass.

Initialization:

$$s_0 = 1, \quad s_1 = 0, \quad t_0 = 0, \quad t_1 = 1$$

Recurrence for $j = 2, 3, \ldots, n$:

$$s_j = s_{j-2} - q_{j-1}s_{j-1}, \quad t_j = t_{j-2} - q_{j-1}t_{j-1}$$

where q_j are the quotients obtained at each division step in the Euclidean algorithm.

Result: At the end,

$$\gcd(a, b) = s_n a + t_n b.$$

Remarks: - This method efficiently computes Bézout coefficients during the gcd calculation. - The proof uses strong induction (see Exercise 44 in Section 5.2 or [Ro10]).

Expressing gcd(252, 198) via Extended Euclidean Algorithm

Euclidean Algorithm Steps:

$$252 = 198 \times 1 + 54 \quad (q_1 = 1)$$

$$198 = 54 \times 3 + 36 \quad (q_2 = 3)$$

$$54 = 36 \times 1 + 18 \quad (q_3 = 1)$$

$$36 = 18 \times 2 + 0 \quad (q_4 = 2)$$

Initialize:

$$s_0 = 1, s_1 = 0, \quad t_0 = 0, t_1 = 1$$

Compute Bézout coefficients:

$$s_2 = s_0 - q_1 s_1 = 1 - 1 \cdot 0 = 1$$

$$t_2 = t_0 - q_1 t_1 = 0 - 1 \cdot 1 = -1$$

$$s_3 = s_1 - q_2 s_2 = 0 - 3 \cdot 1 = -3$$

$$t_3 = t_1 - q_2 t_2 = 1 - 3 \cdot (-1) = 4$$

$$s_4 = s_2 - q_3 s_3 = 1 - 1 \cdot (-3) = 4$$

$$t_4 = t_2 - q_3 t_3 = -1 - 1 \cdot 4 = -5$$

Result:

$$18 = \gcd(252, 198) = 4 \times 252 - 5 \times 198$$

Theorem 7: Cancellation Law for Congruences

Theorem 7: Let m be a positive integer and let a, b, and c be integers. If $ac \equiv bc \pmod{m}$ and $\gcd(c, m) = 1$, then

$$a \equiv b \pmod{m}.$$

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Theorem 7: Let m be a positive integer and let a, b, and c be integers. If $ac \equiv bc \pmod{m}$ and $\gcd(c, m) = 1$, then

$$a \equiv b \pmod{m}.$$

Proof: Since $ac \equiv bc \pmod{m}$, it follows that

$$m \mid ac - bc = c(a - b).$$

By Lemma 2, because $\gcd(c, m) = 1$, we have

$$m \mid (a - b).$$

Therefore,

$$a \equiv b \pmod{m}.$$